

Grade 8
Unit 4
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

Apply the laws of exponents to evaluate each expression. Show your steps in the process.

$$1. \quad \frac{(0.25)^{-2} \cdot (0.25)^5}{(0.25)^3} \quad \frac{0.25^{-2+5}}{0.25^3} = \frac{0.25^3}{0.25^3} = 0.25^{3-3} = 0.25^0 = 1$$

$$2. \quad (-3)^3 \cdot \left(\frac{-5}{3}\right)^{-4} \quad (-3^3) \cdot \left(\frac{3}{-5}\right)^4 = \frac{-27}{1} \cdot \frac{81}{625} = \frac{-2187}{625}$$

For each expression, apply one or more exponent laws to evaluate.

Expression	Value
3. $(3^3)^{-3}$	$3^{3 \cdot -3} = 3^{-9} = \frac{1}{3^9} = \frac{1}{19,683}$
4. $\left(\frac{48}{6}\right)^{-2}$	$\left(\frac{6}{48}\right)^2 = \left(\frac{1}{8}\right)^2 = \frac{1}{64}$
5. $(3^4 \cdot 2^{-4})^{-1}$	$3^{4 \cdot -1} \cdot 2^{-4 \cdot -1} = 3^{-4} \cdot 2^4 = \frac{1}{3^4} \cdot \frac{2^4}{1} = \frac{16}{81}$

Evaluate each expression in the table below. Then identify which exponent law(s) were used to evaluate each expression.

Answer may vary.

Expression	Value	Exponent Law(s)
6. $\frac{6^{-2} \cdot 6^5}{6}$	$\frac{6^{-2+5}}{6^1} = \frac{6^3}{6^1} = 6^2 = 36$	Product of Powers Quotient of Powers
7. $\frac{-9^4}{-9^0 \cdot -9^3}$	$\frac{-9^4}{-9^{0+3}} \cdot \frac{-9^4}{-9^3} = -9^1 = -9$	Product of Powers Quotient of Powers
8. $5^2 \cdot \left(\frac{1}{4}\right)^{-1}$	$5^2 \cdot 4 = 100$	Negative exponent Law
9. $\frac{(-2.5)^0 \cdot (-2.5)^1}{(-2.5)^{-2}}$	$\frac{(-2.5)^{0+1}}{(-2.5)^{-2}} = \frac{-2.5^1}{-2.5^{-2}} = -2.5^{1-(-2)}$ $= (-2.5)^3 = -15.625$	Product of Powers Quotient of Powers
10. $\left(\frac{7}{5}\right)^{-4}$	$\left(\frac{5}{7}\right)^4 = \frac{625}{2,401}$	Negative exponent Law